

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461052.6364 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.02
Transit depth (Rp/Rs)^2	2.54 +/- 0.64 [%]
Orbital Inclination (inc)	86.61 +/- 0.81
Airmass coefficient 1 (a1)	1.797 +/- 0.018
Airmass coefficient 2 (a2)	-0.38 +/- 0.16
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	None
Optimal Aperture	3.71
Optimal Annulus	8.85
Transit Duration (day)	0.093 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.02 vs 0.14195069566578392 (deviation 0.85σ)
Duration fit vs published	0.093 d vs 0.0875 d (deviation 0.34σ)
Mid-time O–C	5.7 min
Depth SNR	8.0
Residual scatter	1.0 %

Light curve

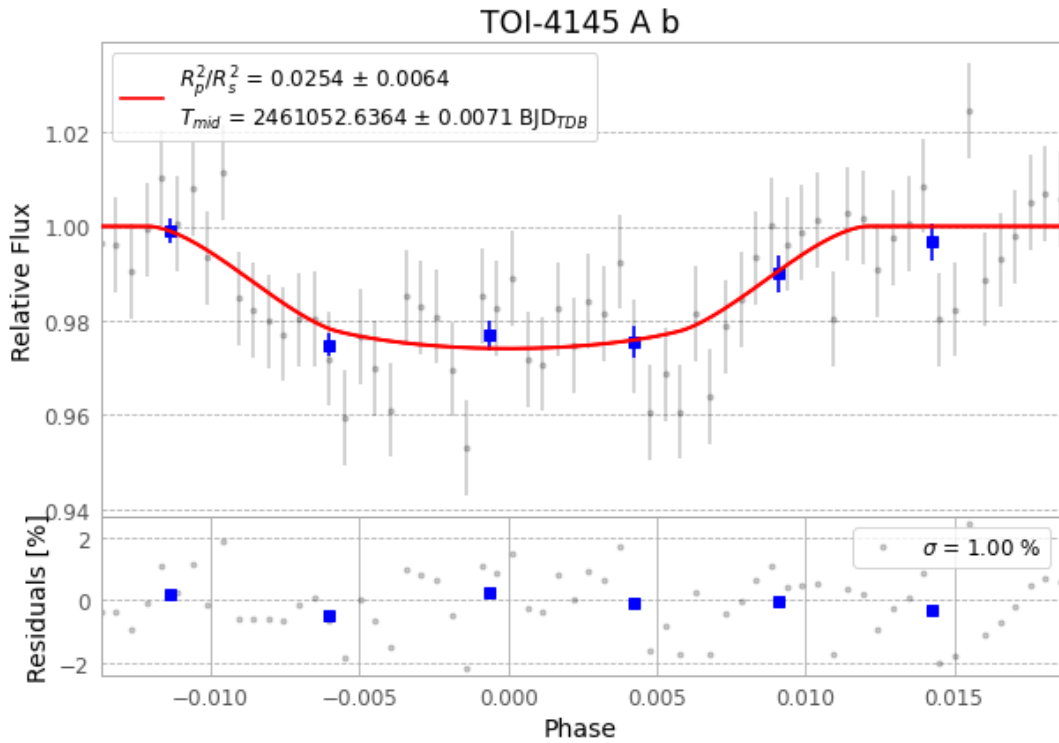


Figure 1 — FinalLightCurve_TOI-4145 A b_2026-01-11.png (SHA-256 d964b18f2a778a7e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 344], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c662505da2476e1e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), TOI4145260112015115.FITS → TOI4145260112050016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4145-260112.log (be6963683f12...), FinalLightCurve_TOI-4145 A b_2026-01-11.png (d964b18f2a77...), FinalLightCurve_TOI-4145 A b_2026-01-11.csv (40f57839d002...)

Compute resources

Wall time 97.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:41Z.